

Qualification Pack



Aircraft Maintenance Technician

QP Code: AAS/Q2009

Version: 1.0

NSQF Level: 4.5

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AAS/Q2009: Aircraft Maintenance Technician

Brief Job Description

The Aircraft Maintenance Technician performs inspection, servicing, troubleshooting, and repair of aircraft systems, structures, and components following regulatory and manufacturer standards. The role includes line maintenance, airframe/powerplant checks, avionics diagnostics, and metal/composite structural repairs using approved procedures. The individual ensures safety, documentation accuracy, and airworthiness while coordinating effectively within a team environment.

Personal Attributes

The individual must be safety-focused, disciplined, and detail-oriented with strong technical accuracy and problem-solving ability. They should demonstrate professionalism, teamwork, clear communication, and the ability to work under operational pressure. Adaptability, willingness to learn new technologies, and maintaining high-quality standards are essential for this role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [AAS/N2025: Work Organization and Managements](#)
2. [AAS/N2026: Communication & Interpersonal Skills](#)
3. [AAS/N2027: Problem Solving, Innovation, & Creativity](#)
4. [AAS/N2028: Sustainable Practices](#)
5. [AAS/N2029: Aircraft and System Inspection Line Maintenance](#)
6. [AAS/N2030: Airframe & Powerplant Component Inspection and Rectification](#)
7. [AAS/N2031: Aircraft Avionic System Inspection and Rectification](#)
8. [AAS/N2032: Aircraft Metal Structure Fabrication and/or Repair](#)
9. [AAS/N2033: Aircraft Composite Inspection and Repair](#)
10. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Aerospace and Aviation
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Sub-Sector	Maintenance Repair and Overhauling
Occupation	Aircraft Maintenance
Country	India
NSQF Level	4.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO/2015/2144.0900
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) OR Completed 3 year diploma after 10th OR 12th grade Pass (or Equivalent) with 1.5 years of experience in airport operation OR 10th grade pass (or Equivalent) with 4 Years of experience in airport opeartion OR Previous relevant Qualification of NSQF Level (4) with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	06/02/2031
NSQC Approval Date	06/02/2026
Version	1.0
Reference code on NQR	QG-4.5-AS-04933-2026-V1-AASSC
NQR Version	1

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AAS/N2025: Work Organization and Managements

Description

This NOS covers the ability to organize, plan, and carry out aircraft maintenance work safely, efficiently, and in accordance with approved procedures. It includes compliance with manuals, safety standards, tool usage, documentation practices, and maintaining airworthiness through disciplined work habits.

Scope

The scope covers the following :

- Application of approved procedures, manuals, and safety standards in all maintenance tasks.
- Proper handling, use, and storage of tools, equipment, and materials.
- Effective planning, workflow management, and maintenance of a clean, safe, and organized work environment.

Elements and Performance Criteria

Follow Safety Standards and Approved Procedures

To be competent, the user/individual on the job must be able to:

- PC1.** Follow the Company Maintenance Policy Manual (MPM) and relevant ATA chapters.
- PC2.** Use approved manuals, manufacturer data, and regulatory documents.
- PC3.** Comply with health, safety, and environmental legislation during maintenance tasks.
- PC4.** Use PPE appropriately including safety footwear, gloves, respirators, and eye/ear protection.
- PC5.** Apply electrostatic discharge (ESD) precautions when handling sensitive systems.
- PC6.** Maintain tool control and ensure proper accountability before and after tasks.

Use Tools, Equipment, and Materials Safely

To be competent, the user/individual on the job must be able to:

- PC7.** Select correct hand tools, power tools, and equipment for required maintenance tasks.
- PC8.** Clean, maintain, and store tools properly to prevent damage or unsafe operation.
- PC9.** Handle materials safely including chemicals, lubricants, sealants, and composites.
- PC10.** Store materials as per organizational and regulatory procedures.
- PC11.** Identify unserviceable tools or equipment and report for repair or replacement.

Plan and Organize Work Effectively

To be competent, the user/individual on the job must be able to:

- PC12.** Plan maintenance work to maximize efficiency and minimize delays.
- PC13.** Prepare the work area and ensure availability of tools and materials prior to tasks.
- PC14.** Maintain a clean, tidy, and FOD-free workspace at all times.
- PC15.** Prioritize tasks based on aircraft schedule, safety requirements, and operational urgency.
- PC16.** Recognize personal authority limits and escalate concerns appropriately.

Maintain Quality, Accuracy, and Airworthiness

To be competent, the user/individual on the job must be able to:

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- PC17.** Measure components accurately using approved measuring tools
- PC18.** Follow the latest revisions of approved manuals and regulated procedures.
- PC19.** Certify completion of tasks only when fully compliant with airworthiness standards.
- PC20.** Maintain high accuracy, attention to detail, and consistency even under pressure.
- PC21.** Document all maintenance actions clearly for traceability and audit requirements.

Work Effectively within a Team Environment

To be competent, the user/individual on the job must be able to:

- PC22.** Collaborate with team members to complete tasks safely and efficiently.
- PC23.** Participate in task planning, shift briefings, and resource allocation.
- PC24.** Support teamwork during time-critical maintenance operations.
- PC25.** Communicate effectively to ensure smooth task execution and safety compliance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company Maintenance Policy Manual (MPM) and its application.
- KU2.** ATA chapters and their use in organizing technical information.
- KU3.** Health, safety, and environmental regulations applicable to aircraft maintenance.
- KU4.** Types and proper use of hand tools, power tools, and machine tools.
- KU5.** Material handling requirements for chemicals, metals, composites, and consumables.
- KU6.** Workflow planning principles and time management in maintenance environments.
- KU7.** Importance of accuracy, documentation, and traceability in aircraft maintenance.
- KU8.** International airworthiness standards and certification protocols.
- KU9.** Consequences of poor work organization or non-compliance.
- KU10.** Roles, responsibilities, and authority boundaries within maintenance teams.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret technical manuals, job cards, and regulatory documents.
- GS2.** Complete documentation accurately using maintenance systems or logbooks.
- GS3.** Communicate clearly with engineers, supervisors, and team members.
- GS4.** Identify workflow inefficiencies and propose practical solutions.
- GS5.** Maintain safety, housekeeping, and FOD control throughout operations.
- GS6.** Use planning and time-management skills to meet maintenance deadlines.
- GS7.** Recognize hazards, report risks, and follow safe work practices.
- GS8.** Work collaboratively with team members to support task completion.
- GS9.** Adapt to changes in procedures, schedules, and revised manuals.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow Safety Standards and Approved Procedures</i>	12	14	-	-
PC1. Follow the Company Maintenance Policy Manual (MPM) and relevant ATA chapters.	2	3	-	-
PC2. Use approved manuals, manufacturer data, and regulatory documents.	2	3	-	-
PC3. Comply with health, safety, and environmental legislation during maintenance tasks.	2	2	-	-
PC4. Use PPE appropriately including safety footwear, gloves, respirators, and eye/ear protection.	2	2	-	-
PC5. Apply electrostatic discharge (ESD) precautions when handling sensitive systems.	2	2	-	-
PC6. Maintain tool control and ensure proper accountability before and after tasks.	2	2	-	-
<i>Use Tools, Equipment, and Materials Safely</i>	10	10	-	-
PC7. Select correct hand tools, power tools, and equipment for required maintenance tasks.	2	2	-	-
PC8. Clean, maintain, and store tools properly to prevent damage or unsafe operation.	2	2	-	-
PC9. Handle materials safely including chemicals, lubricants, sealants, and composites.	2	2	-	-
PC10. Store materials as per organizational and regulatory procedures.	2	2	-	-
PC11. Identify unserviceable tools or equipment and report for repair or replacement.	2	2	-	-
<i>Plan and Organize Work Effectively</i>	10	10	-	-
PC12. Plan maintenance work to maximize efficiency and minimize delays.	2	2	-	-
PC13. Prepare the work area and ensure availability of tools and materials prior to tasks.	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Maintain a clean, tidy, and FOD-free workspace at all times.	2	2	-	-
PC15. Prioritize tasks based on aircraft schedule, safety requirements, and operational urgency.	2	2	-	-
PC16. Recognize personal authority limits and escalate concerns appropriately.	2	2	-	-
<i>Maintain Quality, Accuracy, and Airworthiness</i>	10	10	-	-
PC17. Measure components accurately using approved measuring tools	2	2	-	-
PC18. Follow the latest revisions of approved manuals and regulated procedures.	2	2	-	-
PC19. Certify completion of tasks only when fully compliant with airworthiness standards.	2	2	-	-
PC20. Maintain high accuracy, attention to detail, and consistency even under pressure.	2	2	-	-
PC21. Document all maintenance actions clearly for traceability and audit requirements.	2	2	-	-
<i>Work Effectively within a Team Environment</i>	6	8	-	-
PC22. Collaborate with team members to complete tasks safely and efficiently.	2	2	-	-
PC23. Participate in task planning, shift briefings, and resource allocation.	2	2	-	-
PC24. Support teamwork during time-critical maintenance operations.	1	2	-	-
PC25. Communicate effectively to ensure smooth task execution and safety compliance.	1	2	-	-
NOS Total	48	52	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2025
NOS Name	Work Organization and Managements
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

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AAS/N2026: Communication & Interpersonal Skills

Description

This NOS covers the communication, teamwork, and interpersonal abilities required in an aircraft maintenance environment. It includes accurate technical documentation, professional interactions, teamwork, conflict resolution, and maintaining customer confidence while supporting safe and efficient operations.

Scope

The scope covers the following :

- Accurate and clear communication with engineering teams, flight crew, and support departments.
- Effective teamwork, collaboration, and conflict resolution in high-pressure aviation environments.
- Professional behavior that supports safety, customer confidence, and organizational standards.

Elements and Performance Criteria

Communicate Accurately and Professionally

To be competent, the user/individual on the job must be able to:

- PC1.** Complete technical documentation accurately and legibly.
- PC2.** Use correct technical terminology in reports, logbooks, and defect entries.
- PC3.** Communicate clearly with supervisors, engineers, and supporting staff.
- PC4.** Provide accurate updates on task status, delays, or defects found during maintenance.
- PC5.** Follow communication protocols during aircraft operations and safety-critical tasks.

Maintain Positive Interpersonal Relationships

To be competent, the user/individual on the job must be able to:

- PC6.** Establish and maintain productive working relationships with colleagues and other departments.
- PC7.** Show respect, professionalism, and a positive attitude in all interactions.
- PC8.** Understand the roles and responsibilities of related departments (logistics, stores, engineering).
- PC9.** Support effective coordination during shift handovers and team briefings.
- PC10.** Build customer confidence by demonstrating competence and disciplined behavior.

Demonstrate Effective Teamwork and Collaboration

To be competent, the user/individual on the job must be able to:

- PC11.** Work cooperatively within a team to complete tasks efficiently and safely.
- PC12.** Follow teamwork principles including mutual support, shared responsibility, and clear communication.
- PC13.** Participate actively in team tasks, briefings, and aircraft turnaround operations.
- PC14.** Identify team priorities and adjust work practices to support task completion.
- PC15.** Promote team safety culture by encouraging proper procedures and hazard reporting.

Handle Conflicts and Challenges Professionally

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To be competent, the user/individual on the job must be able to:

- PC16.** Identify misunderstandings and resolve them through professional communication.
- PC17.** Respond constructively to feedback from supervisors or peers.
- PC18.** Address conflicting demands calmly and with a solutions-oriented approach.
- PC19.** Escalate unresolved issues through proper channels when necessary.
- PC20.** Maintain professional behavior even under high workload or time pressure.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of accurate written and verbal communication in aviation safety.
- KU2.** Customer confidence and its role in maintaining organizational reputation.
- KU3.** Roles and responsibilities of related colleagues and team members.
- KU4.** The value of building productive working relationships in the maintenance environment.
- KU5.** Industry-accepted Triple A ,attitude, aptitude, and ability.
- KU6.** Communication techniques for briefing, reporting, and technical discussions.
- KU7.** Basic conflict resolution, feedback handling, and interpersonal communication principles.
- KU8.** Importance of teamwork in completing maintenance tasks efficiently.
- KU9.** Common challenges in communication during live aircraft operations.
- KU10.** Organizational protocols for escalation and information sharing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write clear and accurate technical reports, entries, and documentation.
- GS2.** Read and interpret instructions, manuals, job cards, and organizational policies.
- GS3.** Communicate effectively with colleagues, engineers, and flight crew.
- GS4.** Listen actively and respond appropriately in briefings or team discussions.
- GS5.** Maintain positive behavior and promote a cooperative work environment.
- GS6.** Identify communication gaps and resolve them quickly.
- GS7.** Use problem-solving skills to handle conflicts or misunderstandings.
- GS8.** Adapt communication style based on audience (crew, engineers, support staff).
- GS9.** Support team performance and shared goals through effective collaboration.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate Accurately and Professionally</i>	15	15	-	-
PC1. Complete technical documentation accurately and legibly.	3	3	-	-
PC2. Use corrects technical terminology in reports, logbooks, and defect entries.	3	3	-	-
PC3. Communicate clearly with supervisors, engineers, and supporting staff.	3	3	-	-
PC4. Provide accurate updates on task status, delays, or defects found during maintenance.	3	3	-	-
PC5. Follow communication protocols during aircraft operations and safety-critical tasks.	3	3	-	-
<i>Maintain Positive Interpersonal Relationships</i>	13	15	-	-
PC6. Establish and maintain productive working relationships with colleagues and other departments.	3	3	-	-
PC7. Show respect, professionalism, and a positive attitude in all interactions.	3	3	-	-
PC8. Understand the roles and responsibilities of related departments (logistics, stores, engineering).	3	3	-	-
PC9. Support effective coordination during shift handovers and team briefings.	2	3	-	-
PC10. Build customer confidence by demonstrating competence and disciplined behavior.	2	3	-	-
<i>Demonstrate Effective Teamwork and Collaboration</i>	10	12	-	-
PC11. Work cooperatively within a team to complete tasks efficiently and safely.	2	3	-	-
PC12. Follow teamwork principles including mutual support, shared responsibility, and clear communication.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Participate actively in team tasks, briefings, and aircraft turnaround operations.	2	2	-	-
PC14. Identify team priorities and adjust work practices to support task completion.	2	2	-	-
PC15. Promote team safety culture by encouraging proper procedures and hazard reporting.	2	2	-	-
<i>Handle Conflicts and Challenges Professionally</i>	10	10	-	-
PC16. Identify misunderstandings and resolve them through professional communication.	2	2	-	-
PC17. Respond constructively to feedback from supervisors or peers.	2	2	-	-
PC18. Address conflicting demands calmly and with a solutions-oriented approach.	2	2	-	-
PC19. Escalate unresolved issues through proper channels when necessary.	2	2	-	-
PC20. Maintain professional behavior even under high workload or time pressure.	2	2	-	-
NOS Total	48	52	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2026
NOS Name	Communication & Interpersonal Skills
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

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AAS/N2027: Problem Solving, Innovation, & Creativity

Description

This NOS focuses on the ability to identify, analyze, and resolve technical problems in aircraft maintenance using structured diagnostic approaches. It includes following updated manuals, preventing errors, applying innovative thinking, and ensuring solutions meet airworthiness standards. The individual must demonstrate adaptability, continuous improvement, and accuracy in problem-solving.

Scope

The scope covers the following :

- Identifying and resolving maintenance problems using approved diagnostic methods.
- Applying updated manuals, data revisions, and technical references during troubleshooting.
- Contributing innovative ideas and embracing new technologies within approved practices.

Elements and Performance Criteria

Identify and Analyze Problems

To be competent, the user/individual on the job must be able to:

- PC1.** Identify common maintenance problems such as leaks, electrical faults, corrosion, or system malfunctions.
- PC2.** Gather relevant information from manuals, task cards, and system history.
- PC3.** Distinguish between normal system behavior and abnormal conditions.
- PC4.** Follow step-by-step fault isolation procedures as per the FIM or troubleshooting guide.
- PC5.** Recognize incorrect or conflicting information and report it.

Apply Diagnostic and Troubleshooting Techniques

To be competent, the user/individual on the job must be able to:

- PC6.** Use approved diagnostic tools and equipment to verify system performance.
- PC7.** Interpret readings, measurements, and test results accurately.
- PC8.** Follow the latest revision of the manufacturer's maintenance manuals and documentation.
- PC9.** Perform systematic checks to identify root causes of faults.
- PC10.** Persist in resolving complex or recurring problems.

Prevent Errors and Ensure Work Quality

To be competent, the user/individual on the job must be able to:

- PC11.** Check and verify own work regularly to minimize rework.
- PC12.** Support verification of co-workers' tasks during team operations.
- PC13.** Identify potential issues early and recommend preventive actions.
- PC14.** Ensure all solutions meet international airworthiness standards.
- PC15.** Document problem analysis and corrective actions clearly.

Encourage Innovation and Continuous Improvement

To be competent, the user/individual on the job must be able to:

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- PC16.** Suggest ideas to improve procedures, tools, or workflow.
- PC17.** Adopt new technologies such as digital troubleshooting tools or advanced materials.
- PC18.** Apply creative thinking to propose safer or more efficient solutions.
- PC19.** Take forward innovative ideas to supervisors or management.
- PC20.** Embrace updated methods and approved changes to improve maintenance standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Common problems encountered in mechanical, electrical, avionics, and structural systems.
- KU2.** Importance of following international airworthiness standards in all maintenance tasks.
- KU3.** Diagnostic approaches such as root-cause analysis, elimination method, and systematic checks.
- KU4.** Use of FIM, AMM troubleshooting tables, and other manufacturer documentation.
- KU5.** Latest industry developments: new materials, upgraded systems, advanced diagnostic tools.
- KU6.** Impact of using outdated or incorrect data on aircraft safety.
- KU7.** Importance of cross-checking and verifying own work.
- KU8.** Organizational procedures for reporting problems and proposing improvements.
- KU9.** Safety risks associated with incorrect troubleshooting or poor judgment
- KU10.** Principles of innovation, process improvement, and change management in maintenance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze information logically to identify faults and possible solutions
- GS2.** Interpret technical diagrams, manuals, measurements, and test results.
- GS3.** Record findings and corrective actions accurately.
- GS4.** Communicate effectively during troubleshooting discussions.
- GS5.** Think critically and propose alternative solutions when standard methods fail.
- GS6.** Work collaboratively with others to solve complex issues.
- GS7.** Apply creative thinking to improve maintenance outcomes.
- GS8.** Use digital tools and modern diagnostic equipment effectively.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify and Analyze Problems</i>	15	15	-	-
PC1. Identify common maintenance problems such as leaks, electrical faults, corrosion, or system malfunctions.	3	3	-	-
PC2. Gather relevant information from manuals, task cards, and system history.	3	3	-	-
PC3. Distinguish between normal system behavior and abnormal conditions.	3	3	-	-
PC4. Follow step-by-step fault isolation procedures as per the FIM or troubleshooting guide.	3	3	-	-
PC5. Recognize incorrect or conflicting information and report it.	3	3	-	-
<i>Apply Diagnostic and Troubleshooting Techniques</i>	13	15	-	-
PC6. Use approved diagnostic tools and equipment to verify system performance.	3	3	-	-
PC7. Interpret readings, measurements, and test results accurately.	3	3	-	-
PC8. Follow the latest revision of the manufacturer's maintenance manuals and documentation.	3	3	-	-
PC9. Perform systematic checks to identify root causes of faults.	2	3	-	-
PC10. Persist in resolving complex or recurring problems.	2	3	-	-
<i>Prevent Errors and Ensure Work Quality</i>	10	12	-	-
PC11. Check and verify own work regularly to minimize rework.	2	3	-	-
PC12. Support verification of co-workers' tasks during team operations.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Identify potential issues early and recommend preventive actions.	2	2	-	-
PC14. Ensure all solutions meet international airworthiness standards.	2	2	-	-
PC15. Document problem analysis and corrective actions clearly.	2	2	-	-
<i>Encourage Innovation and Continuous Improvement</i>	10	10	-	-
PC16. Suggest ideas to improve procedures, tools, or workflow.	2	2	-	-
PC17. Adopt new technologies such as digital troubleshooting tools or advanced materials.	2	2	-	-
PC18. Apply creative thinking to propose safer or more efficient solutions.	2	2	-	-
PC19. Take forward innovative ideas to supervisors or management.	2	2	-	-
PC20. Embrace updated methods and approved changes to improve maintenance standards.	2	2	-	-
NOS Total	48	52	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2027
NOS Name	Problem Solving, Innovation, & Creativity
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

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AAS/N2028: Sustainable Practices

Description

This NOS focuses on adopting environmentally responsible and socially inclusive practices within aircraft maintenance. It includes reducing waste, using eco-friendly materials, adopting digital documentation, minimizing fuel and energy consumption, understanding emerging green technologies, and promoting diversity and inclusivity in the workplace.

Scope

The scope covers the following :

- Application of environmentally sustainable maintenance practices.
- Use of digital systems and optimized ground operations to reduce emissions.
- Promotion of inclusivity, diversity, and responsible workplace behavior.

Elements and Performance Criteria

Follow Environmental and Regulatory Standards

To be competent, the user/individual on the job must be able to:

- PC1.** Comply with environmental regulations and organizational sustainability policies.
- PC2.** Use environmentally friendly materials wherever approved and applicable.
- PC3.** Dispose of hazardous waste, chemicals, and consumables safely.
- PC4.** Segregate recyclable items such as metals, plastics, and composite waste
- PC5.** Minimize the use of non-recyclable or harmful materials during maintenance.

Reduce Waste and Promote Sustainable Use of Resources

To be competent, the user/individual on the job must be able to:

- PC6.** Use digital documentation and electronic task cards to reduce paper consumption.
- PC7.** Plan tasks effectively to prevent unnecessary wastage of materials.
- PC8.** Maintain tools and equipment to extend their operational lifespan.
- PC9.** Follow FOD control and good housekeeping practices to reduce environmental impact.
- PC10.** Identify opportunities to conserve resources (materials, electricity, consumables).

Support Environment-Friendly Aircraft Ground Operations

To be competent, the user/individual on the job must be able to:

- PC11.** Minimize usage of the aircraft's auxiliary power unit (APU) to reduce fuel burn.
- PC12.** Operate Ground Support Equipment (GSE) efficiently following eco-friendly procedures.
- PC13.** Recognize the potential of digital technology to optimize fuel and energy usage.
- PC14.** Follow procedures to limit engine ground runs and unnecessary system operation.
- PC15.** Report any environmental hazards or spills immediately.

Understand Emerging Sustainable Aviation Technologies

To be competent, the user/individual on the job must be able to:

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- PC16.** Stay updated on developments such as hydrogen propulsion, battery technologies, HBPR engines, and SAF (Sustainable Aviation Fuel).
- PC17.** Identify how new technologies influence future maintenance practices.
- PC18.** Incorporate knowledge of new materials and methods in maintenance tasks.
- PC19.** Embrace innovative green solutions within approved practices.
- PC20.** Support organizational initiatives toward carbon reduction and sustainability.

Promote Inclusivity and Social Responsibility

To be competent, the user/individual on the job must be able to:

- PC21.** Demonstrate respect for diversity and equal opportunity in the workplace.
- PC22.** Support inclusive teamwork and communication with colleagues from different backgrounds.
- PC23.** Address workplace discrimination or bias by following company policies.
- PC24.** Participate in training and awareness programs related to inclusivity.
- PC25.** Maintain professional behavior that supports an inclusive and ethical work culture.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Environmental regulations and sustainable aviation standards.
- KU2.** Waste disposal procedures for chemicals, oils, composites, and consumables.
- KU3.** Benefits of digital documentation and paper reduction.
- KU4.** Fuel-efficient ground operation practices and their environmental impact.
- KU5.** New aviation technologies: hydrogen propulsion, SAF, hybrid-electric and battery systems.
- KU6.** Organizational policies for diversity, inclusivity, and ethical behavior.
- KU7.** Risks associated with improper waste disposal or pollution.
- KU8.** Workplace practices that contribute to sustainability and resource conservation.
- KU9.** Benefits of eco-friendly materials and energy-efficient tools.
- KU10.** Long-term impact of sustainable operations on aviation safety and the environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow written environmental procedures, sustainability manuals, and safety guidelines.
- GS2.** Communicate effectively regarding environmental risks or non-compliance.
- GS3.** Record and report waste disposal, resource usage, and environmental observations.
- GS4.** Plan tasks to reduce waste and unnecessary resource consumption.
- GS5.** Identify sustainable alternatives in daily maintenance operations.
- GS6.** Use digital tools and systems to support paperless documentation.
- GS7.** Work cooperatively in a diverse and inclusive team environment.
- GS8.** Respond appropriately to environmental emergencies (spills, leakages).
- GS9.** Adopt a proactive attitude towards sustainability and continuous improvement.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow Environmental and Regulatory Standards</i>	10	12	-	-
PC1. Comply with environmental regulations and organizational sustainability policies.	2	3	-	-
PC2. Use environmentally friendly materials wherever approved and applicable.	2	3	-	-
PC3. Dispose of hazardous waste, chemicals, and consumables safely.	2	2	-	-
PC4. Segregate recyclable items such as metals, plastics, and composite waste	2	2	-	-
PC5. Minimize the use of non-recyclable or harmful materials during maintenance.	2	2	-	-
<i>Reduce Waste and Promote Sustainable Use of Resources</i>	10	10	-	-
PC6. Use digital documentation and electronic task cards to reduce paper consumption.	2	2	-	-
PC7. Plan tasks effectively to prevent unnecessary wastage of materials.	2	2	-	-
PC8. Maintain tools and equipment to extend their operational lifespan.	2	2	-	-
PC9. Follow FOD control and good housekeeping practices to reduce environmental impact.	2	2	-	-
PC10. Identify opportunities to conserve resources (materials, electricity, consumables).	2	2	-	-
<i>Support Environment-Friendly Aircraft Ground Operations</i>	10	10	-	-
PC11. Minimize usage of the aircraft's auxiliary power unit (APU) to reduce fuel burn.	2	2	-	-
PC12. Operate Ground Support Equipment (GSE) efficiently following eco-friendly procedures.	2	2	-	-
PC13. Recognize the potential of digital technology to optimize fuel and energy usage.	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Follow procedures to limit engine ground runs and unnecessary system operation.	2	2	-	-
PC15. Report any environmental hazards or spills immediately.	2	2	-	-
<i>Understand Emerging Sustainable Aviation Technologies</i>	10	10	-	-
PC16. Stay updated on developments such as hydrogen propulsion, battery technologies, HBPR engines, and SAF (Sustainable Aviation Fuel).	2	2	-	-
PC17. Identify how new technologies influence future maintenance practices.	2	2	-	-
PC18. Incorporate knowledge of new materials and methods in maintenance tasks.	2	2	-	-
PC19. Embrace innovative green solutions within approved practices.	2	2	-	-
PC20. Support organizational initiatives toward carbon reduction and sustainability.	2	2	-	-
<i>Promote Inclusivity and Social Responsibility</i>	8	10	-	-
PC21. Demonstrate respect for diversity and equal opportunity in the workplace.	2	2	-	-
PC22. Support inclusive teamwork and communication with colleagues from different backgrounds.	2	2	-	-
PC23. Address workplace discrimination or bias by following company policies.	2	2	-	-
PC24. Participate in training and awareness programs related to inclusivity.	1	2	-	-
PC25. Maintain professional behavior that supports an inclusive and ethical work culture.	1	2	-	-
NOS Total	48	52	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2028
NOS Name	Sustainable Practices
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

Qualification Pack

AAS/N2029: Aircraft and System Inspection Line Maintenance

Description

This NOS covers the essential skills and knowledge required to perform aircraft line maintenance activities, including pre-flight, transit, and post-flight inspections. It includes detecting defects, performing operational checks, carrying out basic servicing, completing documentation, complying with regulatory standards, and ensuring the aircraft is safe and serviceable for continued operation.

Scope

The scope covers the following :

- Conducting aircraft walk-around inspections and identifying defects.
- Performing system checks, servicing operations, and basic troubleshooting.
- Completing documentation and following regulatory safety and maintenance standards.

Elements and Performance Criteria

Conduct Aircraft Line Maintenance Inspections

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the purpose and scope of line maintenance in aviation operations.
- PC2.** Perform pre-flight, transit, and post-flight inspections as per AMM procedures.
- PC3.** Identify visible defects such as cracks, corrosion, dents, leaks, and wear.
- PC4.** Inspect tires, brakes, lights, antennas, control surfaces, static ports, and external fittings.
- PC5.** Ensure aircraft servicing panels and access points are secure.
- PC6.** Use proper inspection tools such as flashlights, mirrors, and gauges during checks.

Perform Operational Checks and Basic Troubleshooting

To be competent, the user/individual on the job must be able to:

- PC7.** Carry out operational checks for hydraulics, electrical systems, lighting, APU, and avionics equipment.
- PC8.** Identify abnormal indications, sounds, warnings, or system behavior.
- PC9.** Follow AMM/FIM procedures for initial troubleshooting.
- PC10.** Support engineers in defect rectification activities.
- PC11.** Confirm system functionality after servicing or adjustments.
- PC12.** Apply MEL/CDL procedures to determine aircraft dispatch acceptability.

Perform Basic Servicing and Component Handling

To be competent, the user/individual on the job must be able to:

- PC13.** Check and replenish fluids (engine oil, hydraulic fluid, oxygen, water).
- PC14.** Inspect and service tires, struts, and brake systems.
- PC15.** Replace simple components (bulbs, covers, quick-access items) under guidance.
- PC16.** Ensure safe handling and installation of replaceable units (LRUs).
- PC17.** Maintain cleanliness around critical zones to prevent FOD.

Qualification Pack

Follow Safety and Regulatory Requirements

To be competent, the user/individual on the job must be able to:

- PC18.** Use PPE appropriately while performing line maintenance tasks.
- PC19.** Follow aircraft approach procedures, hazard zones, and ramp safety rules.
- PC20.** Ensure all work complies with DGCA/EASA/FAA line maintenance requirements.
- PC21.** Protect aircraft surfaces and components from damage during inspection.
- PC22.** Apply human factors awareness (fatigue, communication, situational awareness).

Complete Documentation and Reporting

To be competent, the user/individual on the job must be able to:

- PC23.** Record all findings, defects, and servicing actions in aircraft documentation.
- PC24.** Make accurate logbook entries using approved terminology.
- PC25.** Use digital tools for documentation and task card updates.
- PC26.** Report defects immediately to supervisors or certifying staff.
- PC27.** Maintain traceability of actions performed during line checks.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Purpose, scope, and importance of line maintenance in aviation.
- KU2.** Components and operation of major aircraft systems.
- KU3.** AMM, MEL, CDL, FIM, and technical documentation usage.
- KU4.** Types of defects commonly found during walk-around inspections.
- KU5.** Ramp safety rules, marshalling zones, and aircraft hazard areas.
- KU6.** Procedures for performing pre-flight, transit, and post-flight checks.
- KU7.** Basic troubleshooting principles and fault identification.
- KU8.** Regulatory requirements for line maintenance (DGCA/EASA/FAA).
- KU9.** Importance of accurate documentation for airworthiness.
- KU10.** Human factors affecting ground operations (stress, fatigue, teamwork).

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret AMM, MEL/CDL, checklists, and technical orders.
- GS2.** Communicate inspection findings clearly with engineers and flight crew.
- GS3.** Use handheld inspection tools accurately and safely.
- GS4.** Prepare and update digital and manual documentation.
- GS5.** Work efficiently during time-critical turn-around operations.
- GS6.** Identify hazards and follow safe ground handling practices.
- GS7.** Apply critical thinking to identify potential defects early.
- GS8.** Work as part of a team to ensure aircraft readiness.

Qualification Pack

GS9. Manage time and tasks effectively during line maintenance operations.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct Aircraft Line Maintenance Inspections</i>	12	18	-	-
PC1. Understand the purpose and scope of line maintenance in aviation operations.	2	3	-	-
PC2. Perform pre-flight, transit, and post-flight inspections as per AMM procedures.	2	3	-	-
PC3. Identify visible defects such as cracks, corrosion, dents, leaks, and wear.	2	3	-	-
PC4. Inspect tires, brakes, lights, antennas, control surfaces, static ports, and external fittings.	2	3	-	-
PC5. Ensure aircraft servicing panels and access points are secure.	2	3	-	-
PC6. Use proper inspection tools such as flashlights, mirrors, and gauges during checks.	2	3	-	-
<i>Perform Operational Checks and Basic Troubleshooting</i>	12	12	-	-
PC7. Carry out operational checks for hydraulics, electrical systems, lighting, APU, and avionics equipment.	2	2	-	-
PC8. Identify abnormal indications, sounds, warnings, or system behavior.	2	2	-	-
PC9. Follow AMM/FIM procedures for initial troubleshooting.	2	2	-	-
PC10. Support engineers in defect rectification activities.	2	2	-	-
PC11. Confirm system functionality after servicing or adjustments.	2	2	-	-
PC12. Apply MEL/CDL procedures to determine aircraft dispatch acceptability.	2	2	-	-
<i>Perform Basic Servicing and Component Handling</i>	6	10	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Check and replenish fluids (engine oil, hydraulic fluid, oxygen, water).	2	2	-	-
PC14. Inspect and service tires, struts, and brake systems.	1	2	-	-
PC15. Replace simple components (bulbs, covers, quick-access items) under guidance.	1	2	-	-
PC16. Ensure safe handling and installation of replaceable units (LRUs).	1	2	-	-
PC17. Maintain cleanliness around critical zones to prevent FOD.	1	2	-	-
<i>Follow Safety and Regulatory Requirements</i>	5	10	-	-
PC18. Use PPE appropriately while performing line maintenance tasks.	1	2	-	-
PC19. Follow aircraft approach procedures, hazard zones, and ramp safety rules.	1	2	-	-
PC20. Ensure all work complies with DGCA/EASA/FAA line maintenance requirements.	1	2	-	-
PC21. Protect aircraft surfaces and components from damage during inspection.	1	2	-	-
PC22. Apply human factors awareness (fatigue, communication, situational awareness).	1	2	-	-
<i>Complete Documentation and Reporting</i>	5	10	-	-
PC23. Record all findings, defects, and servicing actions in aircraft documentation.	1	2	-	-
PC24. Make accurate logbook entries using approved terminology.	1	2	-	-
PC25. Use digital tools for documentation and task card updates.	1	2	-	-
PC26. Report defects immediately to supervisors or certifying staff.	1	2	-	-
PC27. Maintain traceability of actions performed during line checks.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2029
NOS Name	Aircraft and System Inspection Line Maintenance
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

Qualification Pack

AAS/N2030: Airframe & Powerplant Component Inspection and Rectification

Description

This NOS focuses on the inspection, servicing, troubleshooting, and rectification of airframe structures and powerplant components. It covers identification of defects, performing scheduled maintenance, replacing components, conducting adjustments, and ensuring all airframe and engine-related tasks comply with approved manuals and airworthiness standards.

Scope

The scope covers the following :

- Inspection and defect identification for airframe structures and powerplant systems.
- Servicing, repair, adjustment, and replacement of components using approved procedures.
- Documentation, safety compliance, and verification of airworthiness following rectification

Elements and Performance Criteria

Inspecting Airframe and Powerplant Components

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the structure and function of airframe and powerplant systems.
- PC2.** Conduct inspections on fuselage, wings, control surfaces, landing gear, and engine areas.
- PC3.** Identify corrosion, cracks, wear, leaks, misalignment, distortion, or loose fittings.
- PC4.** Use appropriate tools such as mirrors, borescopes, gauges, and NDT aids.
- PC5.** Follow AMM/EMM instructions for scheduled and unscheduled inspections.
- PC6.** Report all discrepancies promptly and accurately.

Perform Servicing and Component Replacement

To be competent, the user/individual on the job must be able to:

- PC7.** Carry out servicing tasks including lubrication, cleaning, adjustment, and replenishing fluids.
- PC8.** Remove and replace mechanical components such as filters, pumps, valves, actuators, and brackets.
- PC9.** Install new or overhauled components ensuring correct alignment and torque.
- PC10.** Follow proper procedures for sealing, torqueing, locking, and safetying of parts.
- PC11.** Handle components safely to prevent damage or contamination.
- PC12.** Verify system performance after servicing or component installation.

Troubleshoot Airframe and Powerplant Systems

To be competent, the user/individual on the job must be able to:

- PC13.** Identify abnormal sounds, vibrations, fluid leaks, and system behavior.
- PC14.** Use AMM, FIM, CMM, and troubleshooting tables to diagnose faults.
- PC15.** Perform step-by-step fault isolation as per technical data.
- PC16.** Support complex troubleshooting activities under supervision of licensed engineers.

Qualification Pack

PC17. Confirm defect rectification by performing functional tests.

Ensure Compliance with Safety and Airworthiness Standards

To be competent, the user/individual on the job must be able to:

PC18. Use PPE and follow safety protocols during airframe/engine tasks.

PC19. Apply proper procedures for working near engines, fuel systems, and moving components.

PC20. Ensure all tasks comply with DGCA, EASA, FAA maintenance standards.

PC21. Protect airframe surfaces from damage while using tools or equipment.

PC22. Follow correct documentation and traceability requirements for rectification work.

Complete Documentation and Post-Maintenance Checks

To be competent, the user/individual on the job must be able to:

PC23. Record all inspection findings and actions in technical documents.

PC24. Make accurate entries in job cards, logbooks, and maintenance records.

PC25. Verify completed work before releasing systems for next-stage checks.

PC26. Support engineers during post-maintenance system runs or taxi tests.

PC27. Maintain evidence of compliance with approved data and procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Major airframe structures --fuselage, wings, empennage, control surfaces.

KU2. Powerplant systems--intake, compressor, combustion, turbine, exhaust, gearbox.

KU3. AMM, EMM, SRM, CMM, and other technical documents.

KU4. Types of structural and mechanical defects.

KU5. Lubrication systems, cooling systems, fuel systems, and pneumatic/hydraulic interfaces.

KU6. Standard practices for torqueing, safetying, sealing, and fastener installation.

KU7. Use of borescopes, gauges, and precision measuring tools.

KU8. Airworthiness requirements for airframe and engine maintenance.

KU9. Human factors affecting maintenance performance.

KU10. Safety risks associated with engine areas, fuel systems, and moving components.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Read and interpret AMM/EMM/CMM instructions accurately.

GS2. Communicate inspection findings and rectification details with engineers.

GS3. Use measuring and inspection tools confidently.

GS4. Follow technical diagrams and torque charts correctly.

GS5. Maintain proper documentation and traceability.

GS6. Work safely around engines and airframe structures.

GS7. Apply logical thinking during troubleshooting.

Qualification Pack

- GS8.** Collaborate with team members for complex rectification tasks.
- GS9.** Manage time effectively to support scheduled maintenance programs.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspecting Airframe and Powerplant Components</i>	12	18	-	-
PC1. Understand the structure and function of airframe and powerplant systems.	2	3	-	-
PC2. Conduct inspections on fuselage, wings, control surfaces, landing gear, and engine areas.	2	3	-	-
PC3. Identify corrosion, cracks, wear, leaks, misalignment, distortion, or loose fittings.	2	3	-	-
PC4. Use appropriate tools such as mirrors, borescopes, gauges, and NDT aids.	2	3	-	-
PC5. Follow AMM/EMM instructions for scheduled and unscheduled inspections.	2	3	-	-
PC6. Report all discrepancies promptly and accurately.	2	3	-	-
<i>Perform Servicing and Component Replacement</i>	12	11	-	-
PC7. Carry out servicing tasks including lubrication, cleaning, adjustment, and replenishing fluids.	2	1	-	-
PC8. Remove and replace mechanical components such as filters, pumps, valves, actuators, and brackets.	2	2	-	-
PC9. Install new or overhauled components ensuring correct alignment and torque.	2	2	-	-
PC10. Follow proper procedures for sealing, torquing, locking, and safetying of parts.	2	2	-	-
PC11. Handle components safely to prevent damage or contamination.	2	2	-	-
PC12. Verify system performance after servicing or component installation.	2	2	-	-
<i>Troubleshoot Airframe and Powerplant Systems</i>	6	10	-	-
PC13. Identify abnormal sounds, vibrations, fluid leaks, and system behavior.	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Use AMM, FIM, CMM, and troubleshooting tables to diagnose faults.	1	2	-	-
PC15. Perform step-by-step fault isolation as per technical data.	1	2	-	-
PC16. Support complex troubleshooting activities under supervision of licensed engineers.	1	2	-	-
PC17. Confirm defect rectification by performing functional tests.	1	2	-	-
<i>Ensure Compliance with Safety and Airworthiness Standards</i>	5	10	-	-
PC18. Use PPE and follow safety protocols during airframe/engine tasks.	1	2	-	-
PC19. Apply proper procedures for working near engines, fuel systems, and moving components.	1	2	-	-
PC20. Ensure all tasks comply with DGCA, EASA, FAA maintenance standards.	1	2	-	-
PC21. Protect airframe surfaces from damage while using tools or equipment.	1	2	-	-
PC22. Follow correct documentation and traceability requirements for rectification work.	1	2	-	-
<i>Complete Documentation and Post-Maintenance Checks</i>	5	10	-	-
PC23. Record all inspection findings and actions in technical documents.	1	2	-	-
PC24. Make accurate entries in job cards, logbooks, and maintenance records.	1	2	-	-
PC25. Verify completed work before releasing systems for next-stage checks.	1	2	-	-
PC26. Support engineers during post-maintenance system runs or taxi tests.	1	2	-	-
PC27. Maintain evidence of compliance with approved data and procedures.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	59	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2030
NOS Name	Airframe & Powerplant Component Inspection and Rectification
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

Qualification Pack

AAS/N2031: Aircraft Avionic System Inspection and Rectification

Description

This NOS covers the inspection, testing, troubleshooting, and rectification of avionics systems in an aircraft. It includes working with communication, navigation, surveillance, electrical systems, sensors, wiring, and electronic Line Replaceable Units (LRUs). The individual must ensure correct system functionality using approved manuals, wiring diagrams, and safety standards.

Scope

The scope covers the following :

- Inspection and performance verification of avionics and electrical systems.
- Troubleshooting and rectification of wiring, LRUs, connectors, and electronic components.
- Safe handling of avionics equipment and compliance with documentation and ESD protection.

Elements and Performance Criteria

Inspect Avionics and Electrical Systems

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the layout and functions of aircraft avionics systems.
- PC2.** Perform visual checks on wiring, connectors, antennas, sensors, and electrical panels.
- PC3.** Identify defects such as loose connectors, chafed wiring, corrosion, or burnt components.
- PC4.** Inspect avionics bays, equipment racks, and cockpit panels for abnormalities.
- PC5.** Use appropriate tools such as multimeters, testers, and continuity checkers for inspections.
- PC6.** Follow AMM and WDM (Wiring Diagram Manual) guidance for all checks.

Perform Functional Tests and System Checks

To be competent, the user/individual on the job must be able to:

- PC7.** Conduct operational tests on navigation, communication, autoflight, radar, surveillance, and display systems.
- PC8.** Monitor system indicators, BIT (Built-In Test) results, and performance parameters.
- PC9.** Identify abnormal behavior, error codes, or system warnings.
- PC10.** Use built-in diagnostic tools to validate system performance.
- PC11.** Document test results accurately for engineering reference.

Troubleshoot Avionics and Electrical Faults

To be competent, the user/individual on the job must be able to:

- PC12.** Use FIM, WDM, and AMM troubleshooting guides to isolate faults.
- PC13.** Perform continuity checks, insulation resistance tests, and voltage measurements.
- PC14.** Identify faults in LRUs, sensors, wiring bundles, connectors, or circuit breakers.
- PC15.** Carry out step-by-step fault isolation procedures as per approved documents.
- PC16.** Confirm repair effectiveness through re-testing and system verification.

Replace and Install Avionics Components

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC17.** Remove and replace LRUs, modules, sensors, and electronic components safely.
- PC18.** Ensure correct installation, locking, and connection of all units.
- PC19.** Follow ESD protection while handling sensitive avionics parts.
- PC20.** Secure wiring and harnesses using approved installation practices.
- PC21.** Update configuration or perform software loading when required.

Ensure Safety, Compliance, and Documentation

To be competent, the user/individual on the job must be able to:

- PC22.** Use PPE and follow safety precautions for electrical and avionics tasks.
- PC23.** Apply ESD grounding, wrist straps, and protective covers for sensitive equipment.
- PC24.** Follow regulatory standards applicable to avionics maintenance.
- PC25.** Maintain accurate documentation in logs, worksheets, and digital systems.
- PC26.** Report unresolved avionics issues to supervising engineers.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Major avionics systems: communication, navigation, surveillance, autoflight, display, electrical distribution.
- KU2.** Use of Wiring Diagram Manual (WDM), AMM, FIM, and IPC.
- KU3.** Basic electrical principles: voltage, current, resistance, continuity, insulation.
- KU4.** Common avionics defects such as corrosion, wiring chafing, connector damage.
- KU5.** Principles of Built-In Tests (BIT), fault codes, and diagnostic reading.
- KU6.** ESD procedures for handling sensitive electronic equipment.
- KU7.** LRU replacement procedures and installation checks.
- KU8.** Regulatory and airworthiness requirements for avionics work.
- KU9.** Electrical safety hazards and correct mitigation practices.
- KU10.** Communication protocols and reporting in avionics maintenance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret wiring diagrams, fault codes, and system schematics.
- GS2.** Use multimeters, testing tools, and diagnostic equipment effectively.
- GS3.** Communicate clearly with engineering teams and flight crew regarding avionics issues.
- GS4.** Complete technical documentation accurately and promptly.
- GS5.** Work systematically during troubleshooting activities.
- GS6.** Follow ESD protection procedures consistently.
- GS7.** Apply logical thinking and analytical skills for fault isolation.
- GS8.** Collaborate with team members during complex avionics rectification.

Qualification Pack

GS9. Manage time efficiently during system testing and troubleshooting.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect Avionics and Electrical Systems</i>	12	18	-	-
PC1. Understand the layout and functions of aircraft avionics systems.	2	3	-	-
PC2. Perform visual checks on wiring, connectors, antennas, sensors, and electrical panels.	2	3	-	-
PC3. Identify defects such as loose connectors, chafed wiring, corrosion, or burnt components.	2	3	-	-
PC4. Inspect avionics bays, equipment racks, and cockpit panels for abnormalities.	2	3	-	-
PC5. Use appropriate tools such as multimeters, testers, and continuity checkers for inspections.	2	3	-	-
PC6. Follow AMM and WDM (Wiring Diagram Manual) guidance for all checks.	2	3	-	-
<i>Perform Functional Tests and System Checks</i>	10	12	-	-
PC7. Conduct operational tests on navigation, communication, autoflight, radar, surveillance, and display systems.	2	3	-	-
PC8. Monitor system indicators, BIT (Built-In Test) results, and performance parameters.	2	3	-	-
PC9. Identify abnormal behavior, error codes, or system warnings.	2	2	-	-
PC10. Use built-in diagnostic tools to validate system performance.	2	2	-	-
PC11. Document test results accurately for engineering reference.	2	2	-	-
<i>Troubleshoot Avionics and Electrical Faults</i>	8	10	-	-
PC12. Use FIM, WDM, and AMM troubleshooting guides to isolate faults.	2	2	-	-
PC13. Perform continuity checks, insulation resistance tests, and voltage measurements.	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Identify faults in LRUs, sensors, wiring bundles, connectors, or circuit breakers.	2	2	-	-
PC15. Carry out step-by-step fault isolation procedures as per approved documents.	1	2	-	-
PC16. Confirm repair effectiveness through re-testing and system verification.	1	2	-	-
<i>Replace and Install Avionics Components</i>	5	10	-	-
PC17. Remove and replace LRUs, modules, sensors, and electronic components safely.	1	2	-	-
PC18. Ensure correct installation, locking, and connection of all units.	1	2	-	-
PC19. Follow ESD protection while handling sensitive avionics parts.	1	2	-	-
PC20. Secure wiring and harnesses using approved installation practices.	1	2	-	-
PC21. Update configuration or perform software loading when required.	1	2	-	-
<i>Ensure Safety, Compliance, and Documentation</i>	5	10	-	-
PC22. Use PPE and follow safety precautions for electrical and avionics tasks.	1	2	-	-
PC23. Apply ESD grounding, wrist straps, and protective covers for sensitive equipment.	1	2	-	-
PC24. Follow regulatory standards applicable to avionics maintenance.	1	2	-	-
PC25. Maintain accurate documentation in logs, worksheets, and digital systems.	1	2	-	-
PC26. Report unresolved avionics issues to supervising engineers.	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2031
NOS Name	Aircraft Avionic System Inspection and Rectification
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

Qualification Pack

AAS/N2032: Aircraft Metal Structure Fabrication and/or Repair

Description

This NOS covers the skills required to inspect, fabricate, and repair metal structures on an aircraft. It includes working with sheet metal, structural components, fasteners, and metallic assemblies using approved manuals and tools. Tasks involve measuring, cutting, drilling, forming, riveting, and ensuring structural integrity while maintaining airworthiness and safety standards.

Scope

The scope covers the following :

- Inspection and assessment of metal structures for defects or damage.
- Fabrication, repair, and assembly of sheet metal components using approved practices.
- Ensuring airworthiness, accuracy, alignment, and documentation during metal repairs.

Elements and Performance Criteria

Inspect Metal Structures and Components

To be competent, the user/individual on the job must be able to:

- PC1.** Understand aircraft structural zones, load paths, and metal assemblies.
- PC2.** Conduct visual inspections for corrosion, cracks, dents, deformation, and loose fasteners.
- PC3.** Identify fatigue damage, metal distortion, and fastener failures.
- PC4.** Use inspection tools such as straight edges, feeler gauges, and magnifiers.
- PC5.** Follow SRM (Structural Repair Manual) for inspection procedures.
- PC6.** Document all findings accurately.

Perform Sheet Metal Fabrication

To be competent, the user/individual on the job must be able to:

- PC7.** Measure materials accurately using rulers, calipers, and templates.
- PC8.** Mark and layout workpieces based on repair instructions.
- PC9.** Cut, shear, or saw sheet metal to required dimensions.
- PC10.** Bend and form metal using brakes, rollers, or forming tools.
- PC11.** Deburr, clean, and prepare surfaces before assembly.
- PC12.** Follow standard practices for material preparation and forming.

Install Fasteners and Structural Attachments

To be competent, the user/individual on the job must be able to:

- PC13.** Drill holes to precise size and alignment using approved methods.
- PC14.** Install rivets, bolts, screws, collars, and special fasteners correctly.
- PC15.** Set rivets using squeeze tools, rivet guns, or bucking bars.
- PC16.** Perform fastener removal and replacement safely.
- PC17.** Check fastener grip length, torque values, and locking requirements.
- PC18.** Ensure assembled structures meet alignment and fit requirements.

Qualification Pack

Perform Structural Repairs Using Approved Data

To be competent, the user/individual on the job must be able to:

- PC19.** Interpret SRM repair instructions accurately.
- PC20.** Fabricate repair doublers, patches, stiffeners, and brackets as specified.
- PC21.** Position and secure repair components using clamps and fixtures.
- PC22.** Complete repairs ensuring correct edge distance, pitch, and structural strength.
- PC23.** Perform quality checks on completed repairs before closing up panels.
- PC24.** Ensure all repairs comply with airworthiness standards.

Follow Safety and Documentation Requirements

To be competent, the user/individual on the job must be able to:

- PC25.** Use PPE and follow safety requirements for cutting, drilling, and riveting.
- PC26.** Prevent FOD and protect aircraft surfaces during metal work.
- PC27.** Maintain tools and equipment to ensure safe operation.
- PC28.** Complete documentation for repairs, measurements, and material usage.
- PC29.** Escalate structural defects outside scope to licensed engineers.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Aircraft structural design basics and load-bearing components.
- KU2.** Types of metals used in aviation: aluminium alloys, titanium, steel.
- KU3.** Common structural defects: corrosion, cracking, buckling, rivet failure.
- KU4.** SRM, AMM, and standard aircraft metal repair practices.
- KU5.** Use of sheet metal tools: snips, brakes, rivet guns, drills, deburring tools.
- KU6.** Principles of edge distance, pitch, alignment, and fastener selection.
- KU7.** Types of fasteners: solid rivets, blind rivets, Hi-Lok, Jo-Bolt, NAS bolts.
- KU8.** Heat treatment, stress considerations, and material properties.
- KU9.** Safety hazards during metalwork (sharp edges, noise, debris).
- KU10.** Airworthiness requirements for structural repair and documentation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret SRM drawings, repair schemes, and layout diagrams.
- GS2.** Measure accurately using calipers, micrometers, and templates.
- GS3.** Communicate repair progress and findings with engineers and inspectors.
- GS4.** Maintain tool safety and housekeeping in metal workshops.
- GS5.** Apply problem-solving skills for fitment issues or alignment challenges.
- GS6.** Work with precision under time constraints.
- GS7.** Keep accurate records of repair materials and components.

Qualification Pack

- GS8.** Coordinate effectively with team members during fabrication tasks.
- GS9.** Follow quality and airworthiness standards consistently.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect Metal Structures and Components</i>	12	14	-	-
PC1. Understand aircraft structural zones, load paths, and metal assemblies.	2	3	-	-
PC2. Conduct visual inspections for corrosion, cracks, dents, deformation, and loose fasteners.	2	3	-	-
PC3. Identify fatigue damage, metal distortion, and fastener failures.	2	2	-	-
PC4. Use inspection tools such as straight edges, feeler gauges, and magnifiers.	2	2	-	-
PC5. Follow SRM (Structural Repair Manual) for inspection procedures.	2	2	-	-
PC6. Document all findings accurately.	2	2	-	-
<i>Perform Sheet Metal Fabrication</i>	11	12	-	-
PC7. Measure materials accurately using rulers, calipers, and templates.	2	2	-	-
PC8. Mark and layout workpieces based on repair instructions.	2	2	-	-
PC9. Cut, shear, or saw sheet metal to required dimensions.	2	2	-	-
PC10. Bend and form metal using brakes, rollers, or forming tools.	2	2	-	-
PC11. Deburr, clean, and prepare surfaces before assembly.	2	2	-	-
PC12. Follow standard practices for material preparation and forming.	1	2	-	-
<i>Install Fasteners and Structural Attachments</i>	6	12	-	-
PC13. Drill holes to precise size and alignment using approved methods.	1	2	-	-
PC14. Install rivets, bolts, screws, collars, and special fasteners correctly.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Set rivets using squeeze tools, rivet guns, or bucking bars.	1	2	-	-
PC16. Perform fastener removal and replacement safely.	1	2	-	-
PC17. Check fastener grip length, torque values, and locking requirements.	1	2	-	-
PC18. Ensure assembled structures meet alignment and fit requirements.	1	2	-	-
<i>Perform Structural Repairs Using Approved Data</i>	6	12	-	-
PC19. Interpret SRM repair instructions accurately.	1	2	-	-
PC20. Fabricate repair doublers, patches, stiffeners, and brackets as specified.	1	2	-	-
PC21. Position and secure repair components using clamps and fixtures.	1	2	-	-
PC22. Complete repairs ensuring correct edge distance, pitch, and structural strength.	1	2	-	-
PC23. Perform quality checks on completed repairs before closing up panels.	1	2	-	-
PC24. Ensure all repairs comply with airworthiness standards.	1	2	-	-
<i>Follow Safety and Documentation Requirements</i>	5	10	-	-
PC25. Use PPE and follow safety requirements for cutting, drilling, and riveting.	1	2	-	-
PC26. Prevent FOD and protect aircraft surfaces during metal work.	1	2	-	-
PC27. Maintain tools and equipment to ensure safe operation.	1	2	-	-
PC28. Complete documentation for repairs, measurements, and material usage.	1	2	-	-
PC29. Escalate structural defects outside scope to licensed engineers.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2032
NOS Name	Aircraft Metal Structure Fabrication and/or Repair
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

Qualification Pack

AAS/N2033: Aircraft Composite Inspection and Repair

Description

This NOS covers the competencies required to inspect, evaluate, and repair composite aircraft structures. It includes identifying damage, preparing repair areas, performing layup and vacuum-bagging processes, curing composite materials, and ensuring repairs meet structural and airworthiness standards. The individual must follow approved manuals, material handling practices, and safety procedures.

Scope

The scope covers the following :

- Inspection and evaluation of composite components for structural integrity.
- Performing composite repair procedures, including layup, curing, and finishing.
- Compliance with SRM instructions, safety standards, and documentation requirements.

Elements and Performance Criteria

Inspect Composite Components and Structures

To be competent, the user/individual on the job must be able to:

- PC1.** Identify composite materials such as carbon fiber, fiberglass, Kevlar, and honeycomb structures.
- PC2.** Conduct visual and tap-test inspections to detect delamination, impact damage, cracks, moisture ingress, and core separation.
- PC3.** Identify dents, abrasions, resin starvation, and fiber breakage.
- PC4.** Use inspection tools such as flashlights, straight edges, moisture meters, and NDI aids.
- PC5.** Follow SRM procedures for composite inspection and classification of damage.
- PC6.** Record all findings accurately for engineering review.

Prepare Repair Area and Materials

To be competent, the user/individual on the job must be able to:

- PC7.** Clean and prepare damaged areas following approved procedures.
- PC8.** Remove loose or damaged material safely using trimming and sanding tools.
- PC9.** Prepare scarf or stepped joints as specified in the repair manual.
- PC10.** Lay out and cut repair plies according to direction, orientation, and SRM ply schedule.
- PC11.** Select appropriate resin systems, adhesives, and core materials.
- PC12.** Ensure correct environmental conditions for composite repair (temperature, humidity).

Perform Composite Layup and Vacuum-Bagging

To be competent, the user/individual on the job must be able to:

- PC13.** Arrange composite plies in the correct sequence and fiber orientation.
- PC14.** Apply resin or adhesive uniformly to plies and repair surfaces.
- PC15.** Assemble vacuum-bagging materials such as release film, breather cloth, sealant tape, and vacuum bags.
- PC16.** Operate vacuum pumps and confirm proper vacuum levels before curing.

Qualification Pack

PC17. Ensure layup is contamination-free and properly aligned.

Complete Curing, Finishing, and Quality Checks

To be competent, the user/individual on the job must be able to:

PC18. Cure composite repairs using heat blankets, ovens, or controlled curing systems.

PC19. Monitor curing temperature and duration as per manufacturer specifications.

PC20. Trim, sand, and finish cured repairs as required.

PC21. Perform post-cure inspection to ensure structural integrity.

PC22. Verify completed repair meets SRM tolerances and airworthiness requirements.

PC23. Protect surrounding areas and prevent FOD during repair operations.

Follow Safety, Handling, and Documentation Requirements

To be competent, the user/individual on the job must be able to:

PC24. Use PPE appropriate for composite materials (dust masks, goggles, gloves).

PC25. Handle chemicals, resins, and adhesives safely.

PC26. Dispose of composite waste, dust, and hazardous materials responsibly.

PC27. Maintain accurate documentation of repair materials, procedures, and results.

PC28. Report non-conforming conditions to engineering staff immediately.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Types of composite materials used in aircraft structures.

KU2. Properties of carbon, glass, aramid fibers, and core materials.

KU3. Typical composite defects: delamination, disbonding, impact damage, moisture ingress.

KU4. Inspection techniques: visual checks, tap-tests, moisture measurement, NDI.

KU5. SRM instructions for composite repair methods and ply schedules.

KU6. Layup techniques, fiber orientation, and resin application methods.

KU7. Vacuum-bagging principles and curing requirements.

KU8. Safety considerations when handling composites, adhesives, and chemicals.

KU9. Environmental conditions affecting composite quality.

KU10. Documentation and airworthiness standards for composite repair

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Read and interpret SRM, AMM, and composite repair drawings.

GS2. Use cutting tools, sanding tools, and vacuum-bagging equipment safely.

GS3. Communicate repair progress and findings with engineers and quality inspectors.

GS4. Maintain cleanliness and eliminate contamination during composite work.

GS5. Apply problem-solving skills during layup and curing issues.

GS6. Work precisely to meet ply orientation, thickness, and dimensional accuracy.

Qualification Pack

- GS7.** Control time and environmental factors during composite repair processes.
- GS8.** Work collaboratively with team members in composite workshops.
- GS9.** Maintain accurate records of materials, curing cycles, and repair details.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect Composite Components and Structures</i>	12	16	-	-
PC1. Identify composite materials such as carbon fiber, fiberglass, Kevlar, and honeycomb structures.	2	3	-	-
PC2. Conduct visual and tap-test inspections to detect delamination, impact damage, cracks, moisture ingress, and core separation.	2	3	-	-
PC3. Identify dents, abrasions, resin starvation, and fiber breakage.	2	3	-	-
PC4. Use inspection tools such as flashlights, straight edges, moisture meters, and NDI aids.	2	3	-	-
PC5. Follow SRM procedures for composite inspection and classification of damage.	2	2	-	-
PC6. Record all findings accurately for engineering review.	2	2	-	-
<i>Prepare Repair Area and Materials</i>	12	12	-	-
PC7. Clean and prepare damaged areas following approved procedures.	2	2	-	-
PC8. Remove loose or damaged material safely using trimming and sanding tools.	2	2	-	-
PC9. Prepare scarf or stepped joints as specified in the repair manual.	2	2	-	-
PC10. Lay out and cut repair plies according to direction, orientation, and SRM ply schedule.	2	2	-	-
PC11. Select appropriate resin systems, adhesives, and core materials.	2	2	-	-
PC12. Ensure correct environmental conditions for composite repair (temperature, humidity).	2	2	-	-
<i>Perform Composite Layup and Vacuum-Bagging</i>	5	10	-	-
PC13. Arrange composite plies in the correct sequence and fiber orientation.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Apply resin or adhesive uniformly to plies and repair surfaces.	1	2	-	-
PC15. Assemble vacuum-bagging materials such as release film, breather cloth, sealant tape, and vacuum bags.	1	2	-	-
PC16. Operate vacuum pumps and confirm proper vacuum levels before curing.	1	2	-	-
PC17. Ensure layup is contamination-free and properly aligned.	1	2	-	-
<i>Complete Curing, Finishing, and Quality Checks</i>	6	12	-	-
PC18. Cure composite repairs using heat blankets, ovens, or controlled curing systems.	1	2	-	-
PC19. Monitor curing temperature and duration as per manufacturer specifications.	1	2	-	-
PC20. Trim, sand, and finish cured repairs as required.	1	2	-	-
PC21. Perform post-cure inspection to ensure structural integrity.	1	2	-	-
PC22. Verify completed repair meets SRM tolerances and airworthiness requirements.	1	2	-	-
PC23. Protect surrounding areas and prevent FOD during repair operations.	1	2	-	-
<i>Follow Safety, Handling, and Documentation Requirements</i>	5	10	-	-
PC24. Use PPE appropriate for composite materials (dust masks, goggles, gloves).	1	2	-	-
PC25. Handle chemicals, resins, and adhesives safely.	1	2	-	-
PC26. Dispose of composite waste, dust, and hazardous materials responsibly.	1	2	-	-
PC27. Maintain accurate documentation of repair materials, procedures, and results.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Report non-conforming conditions to engineering staff immediately.	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N2033
NOS Name	Aircraft Composite Inspection and Repair
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Maintenance
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2031
NSQC Clearance Date	06/02/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- 1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
- 2.The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
- 3.Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
- 4.Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
- 5.Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
- 6.To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
- 7.In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
AAS/N2025.Work Organization and Managements	48	52	0	0	100	5
AAS/N2026.Communication & Interpersonal Skills	48	52	0	0	100	5
AAS/N2027.Problem Solving, Innovation, & Creativity	48	52	0	0	100	5
AAS/N2028.Sustainable Practices	48	52	0	0	100	5
AAS/N2029.Aircraft and System Inspection Line Maintenance	40	60	0	0	100	15
AAS/N2030.Airframe & Powerplant Component Inspection and Rectification	40	59	0	0	99	15
AAS/N2031.Aircraft Avionic System Inspection and Rectification	40	60	0	0	100	15
AAS/N2032.Aircraft Metal Structure Fabrication and/or Repair	40	60	0	0	100	15
AAS/N2033.Aircraft Composite Inspection and Repair	40	60	0	0	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	412	537	-	-	949	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.